39. Config Automation Mindset (Ansible-oriented)

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Phase: Phase 11 — Containers and cloud Linux
Estimated time: 1–2 days
Slug: `39-automation-ansible`

Learning objectives

- Prefer idempotent config management
- Separate inventory from playbooks conceptually
- Avoid snowflake servers
- Combine with immutable images when appropriate

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

Ansible (and friends) apply desired state over SSH.

Principles beat tool trivia:

- Idempotent tasks
- Versioned config
- Small blast radius
- Golden images + automation hybrid

Snowflake servers that nobody can recreate are career-limiting in infra.

Common mistakes

- Manual SSH drift with no record
- One mega playbook without checks
- Automating before understanding

Exercises

Work in `exercises/39-automation-ansible/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/39-automation-ansible/` until you have an honest failing attempt.

Self-check

1. What is idempotency?
2. Why avoid snowflakes?
3. SSH role in Ansible?

Further reading

- https://docs.ansible.com/